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Jai-Kumar786 05-16-2025 Understanding data on A123 Battery

29ca257 · 1 minute ago



8282 lines (8282 loc) · 307 KB

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LOW CURRENT OCV AT -10 degree celcius

In [1]:

```
import pandas as pd

# Read a specific sheet by name
low_curr_ocv_minus_10= pd.read_excel("A1-007-OCV-10-20120629.xlsx", sheet_n

low_curr_ocv_minus_10.head(10)
```

Out[1]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
0	1	3.010944	2012-06-29 16:51:38	3.010944	1	1	0.
1	2	3.026961	2012-06-29 16:51:38	0.015765	2	1	1.
2	3	8.035992	2012-06-29 16:51:44	5.007575	3	1	0.
3	4	13.043648	2012-06-29 16:51:49	10.015231	3	1	0.
4	5	18.051127	2012-06-29 16:51:54	15.022710	3	1	0.
5	6	23.058702	2012-06-29 16:51:59	20.030285	3	1	0.
6	7	28.066285	2012-06-29 16:52:04	25.037868	3	1	0.
7	8	33.073845	2012-06-29 16:52:09	30.045427	3	1	0.
8	9	38.081417	2012-06-29 16:52:14	35.053000	3	1	0.
9	10	43.089009	2012-06-29 16:52:19	40.060591	3	1	0.

In [2]:

```
low_curr_ocv_minus_10.describe()
```

Out[2]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	29785.00000	29785.000000	29785	29785.000000	29785.000000
mean	14893.00000	79452.354943	2012-06-30 14:55:48.690951936	36718.816502	5.960987
min	1.00000	3.010944	2012-06-29 16:51:38	0.015765	1.000000
25%	7447.00000	40580.632806	2012-06-30 04:07:57	18022.251663	5.000000
50%	14893.00000	77867.013967	2012-06-30 14:29:23	36665.442255	5.000000
75%	22339.00000	118452.093694	2012-07-01 01:45:49	55308.632827	7.000000
max	29785.00000	159037.139119	2012-07-01 13:02:14	76334.569682	8.000000
std	8598.33322	44515.946548	NaN	21570.964248	1.013849

In [3]:

```
low_curr_ocv_minus_10.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29785 entries, 0 to 29784
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            29785 non-null  int64
1   Test_Time(s)                          29785 non-null  float64
2   Date_Time                             29785 non-null  datetime64[ns]
3   Step_Time(s)                          29785 non-null  float64
4   Step_Index                            29785 non-null  int64
5   Cycle_Index                           29785 non-null  int64
6   Current(A)                            29785 non-null  float64
7   Voltage(V)                            29785 non-null  float64
8   Charge_Capacity(Ah)                   29785 non-null  float64
9   Discharge_Capacity(Ah)                29785 non-null  float64
10  Charge_Energy(Wh)                     29785 non-null  float64
11  Discharge_Energy(Wh)                  29785 non-null  float64
12  dV/dt(V/s)                            29785 non-null  float64
13  Internal_Resistance(Ohm)               29785 non-null  int64
14  Is_FC_Data                            29785 non-null  int64
15  AC_Impedance(Ohm)                     29785 non-null  int64
16  ACI_Phase_Angle(Deg)                  29785 non-null  int64
17  Temperature (C)_1                      29785 non-null  float64
18  Temperature (C)_2                      29785 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.3 MB
```

LOW CURRENT OCV AT 0 degree celcius

In [5]:

```
low_curr_ocv_0=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A
```

In [6]:

```
low_curr_ocv_0.head(10)
```

Out[6]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
0	1	3.014942	2012-06-18 11:49:18	3.014942	1	1	0.
1	2	3.046301	2012-06-18 11:49:18	0.031089	2	1	1.
2	3	8.054232	2012-06-18 11:49:23	5.007584	3	1	0.
3	4	13.061941	2012-06-18 11:49:28	10.015293	3	1	0.
4	5	18.073395	2012-06-18 11:49:33	15.026747	3	1	0.
5	6	23.080863	2012-06-18 11:49:38	20.034215	3	1	0.
6	7	28.088504	2012-06-18 11:49:43	25.041856	3	1	0.
7	8	33.096150	2012-06-18 11:49:48	30.049501	3	1	0.
8	9	38.103670	2012-06-18 11:49:53	35.057022	3	1	0.
9	10	43.111231	2012-06-18 11:49:58	40.064583	3	1	0.

In [7]:

```
low_curr_ocv_0.describe()
```

Out[7]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	30249.000000	30249.000000	30249	30249.000000	30249.000000
mean	15125.000000	80615.643879	2012-06-19 10:12:51.466792448	36962.274515	5.962048
min	1.000000	3.014942	2012-06-18 11:49:18	0.031089	1.000000

25%	7563.000000	41162.668955	2012-06-18 23:15:18	17997.213788	5.000000
50%	15125.000000	79029.928468	2012-06-19 09:46:26	36930.843545	5.000000
75%	22687.000000	120193.073943	2012-06-19 21:12:29	55864.473308	7.000000
max	30249.000000	161359.267642	2012-06-20 08:38:36	76186.533244	8.000000
std	8732.278483	45209.697396	NaN	21853.017650	1.030102

In [8]: `low_curr_ocv_0.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30249 entries, 0 to 30248
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            30249 non-null  int64
1   Test_Time(s)                          30249 non-null  float64
2   Date_Time                             30249 non-null  datetime64[ns]
3   Step_Time(s)                          30249 non-null  float64
4   Step_Index                            30249 non-null  int64
5   Cycle_Index                           30249 non-null  int64
6   Current(A)                            30249 non-null  float64
7   Voltage(V)                            30249 non-null  float64
8   Charge_Capacity(Ah)                   30249 non-null  float64
9   Discharge_Capacity(Ah)                 30249 non-null  float64
10  Charge_Energy(Wh)                      30249 non-null  float64
11  Discharge_Energy(Wh)                   30249 non-null  float64
12  dV/dt(V/s)                             30249 non-null  float64
13  Internal_Resistance(Ohm)                30249 non-null  int64
14  Is_FC_Data                             30249 non-null  int64
15  AC_Impedance(Ohm)                      30249 non-null  int64
16  ACI_Phase_Angle(Deg)                   30249 non-null  int64
17  Temperature (C)_1                      30249 non-null  float64
18  Temperature (C)_2                      30249 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.4 MB
```

LOW CURRENT OCV AT 10 degree celcius

In [10]: `low_curr_ocv_10=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\`

In [11]: `low_curr_ocv_10.head(10)`

Out[11]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
			2012-06-				
0	1	3.014079	11	3.014079	1	1	0

			18:47:47			
1	2	3.025053	2012-06-11 18:47:47	0.000002	2	1 1.
2	3	8.032849	2012-06-11 18:47:52	5.007554	3	1 0.
3	4	13.040499	2012-06-11 18:47:57	10.015204	3	1 0.
4	5	18.046321	2012-06-11 18:48:02	15.021026	3	1 0.
5	6	23.053825	2012-06-11 18:48:07	20.028530	3	1 0.
6	7	28.061527	2012-06-11 18:48:12	25.036233	3	1 0.
7	8	33.068989	2012-06-11 18:48:17	30.043694	3	1 0.
8	9	38.076571	2012-06-11 18:48:22	35.051276	3	1 0.
9	10	43.084085	2012-06-11 18:48:27	40.058790	3	1 0.



In [12]: `low_curr_ocv_10.describe()`

Out[12]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	31898.00000	31898.000000	31898	31898.000000	31898.000000
mean	15949.50000	84407.542774	2012-06-12 23:21:28.604269824	35034.642104	5.807605
min	1.00000	3.014079	2012-06-11 18:47:47	0.000002	1.000000
25%	7975.25000	43127.482188	2012-06-12 12:56:42.249999872	14612.052933	5.000000
50%	15949.50000	83059.113340	2012-06-13 00:02:14.500000	34527.490563	5.000000
75%	23923.75000	126287.349589	2012-06-13 12:02:42.750000128	54492.820953	7.000000
max	31898.00000	169513.642825	2012-06-14 00:03:10	75041.088878	8.000000

std 9208.30378 47856.153587 NaN 22318.590732 1.210257

In [13]: `low_curr_ocv_10.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31898 entries, 0 to 31897
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            31898 non-null  int64
1   Test_Time(s)                          31898 non-null  float64
2   Date_Time                             31898 non-null  datetime64[ns]
3   Step_Time(s)                          31898 non-null  float64
4   Step_Index                            31898 non-null  int64
5   Cycle_Index                           31898 non-null  int64
6   Current(A)                            31898 non-null  float64
7   Voltage(V)                            31898 non-null  float64
8   Charge_Capacity(Ah)                   31898 non-null  float64
9   Discharge_Capacity(Ah)                31898 non-null  float64
10  Charge_Energy(Wh)                     31898 non-null  float64
11  Discharge_Energy(Wh)                  31898 non-null  float64
12  dV/dt(V/s)                           31898 non-null  float64
13  Internal_Resistance(Ohm)               31898 non-null  int64
14  Is_FC_Data                            31898 non-null  int64
15  AC_Impedance(Ohm)                     31898 non-null  int64
16  ACI_Phase_Angle(Deg)                  31898 non-null  int64
17  Temperature (C)_1                     31898 non-null  float64
18  Temperature (C)_2                     31898 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.6 MB
```

LOW CURRENT OCV AT 20 degree celcius

In [14]: `low_curr_ocv_20=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\`

In [15]: `low_curr_ocv_20.head(10)`

Out[15]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
0	1	3.007810	2012-06-14 12:13:15	3.007810	1	1	0.
1	2	3.008080	2012-06-14 12:13:15	0.000002	2	1	1.
2	3	8.015990	2012-06-14 12:13:20	5.007662	3	1	0.
3	4	13.023481	2012-06-14 12:13:25	10.015153	3	1	0.

12:13:25

4	5	18.035039	2012-06-14 12:13:30	15.026711	3	1	0
5	6	23.042624	2012-06-14 12:13:35	20.034296	3	1	0
6	7	28.050181	2012-06-14 12:13:40	25.041853	3	1	0
7	8	33.057714	2012-06-14 12:13:45	30.049386	3	1	0
8	9	38.065421	2012-06-14 12:13:50	35.057093	3	1	0
9	10	43.072973	2012-06-14 12:13:55	40.064645	3	1	0

In [16]: `low_curr_ocv_20.describe()`

Out[16]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	31018.000000	31018.000000	31018	31018.000000	31018.000000
mean	15509.500000	82530.938666	2012-06-15 11:08:43.815655168	37436.752180	5.955671
min	1.000000	3.007810	2012-06-14 12:13:15	0.000002	1.000000
25%	7755.250000	42125.027738	2012-06-14 23:55:17.249999872	17983.441759	5.000000
50%	15509.500000	80954.993049	2012-06-15 10:42:27.500000	37399.049136	5.000000
75%	23263.750000	123082.373990	2012-06-15 22:24:35.750000128	56814.656519	7.000000
max	31018.000000	165211.592285	2012-06-16 10:06:45	76708.858391	8.000000
std	8954.269661	46356.820098	NaN	22362.338814	1.053492

In [17]: `low_curr_ocv_20.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31018 entries, 0 to 31017
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
```



```

0   Data_Point          31018 non-null   int64
1   Test_Time(s)        31018 non-null   float64
2   Date_Time           31018 non-null   datetime64[ns]
3   Step_Time(s)        31018 non-null   float64
4   Step_Index          31018 non-null   int64
5   Cycle_Index         31018 non-null   int64
6   Current(A)          31018 non-null   float64
7   Voltage(V)          31018 non-null   float64
8   Charge_Capacity(Ah) 31018 non-null   float64
9   Discharge_Capacity(Ah) 31018 non-null   float64
10  Charge_Energy(Wh)    31018 non-null   float64
11  Discharge_Energy(Wh) 31018 non-null   float64
12  dV/dt(V/s)          31018 non-null   float64
13  Internal_Resistance(Ohm) 31018 non-null   int64
14  Is_FC_Data          31018 non-null   int64
15  AC_Impedance(Ohm)    31018 non-null   int64
16  ACI_Phase_Angle(Deg) 31018 non-null   int64
17  Temperature (C)_1    31018 non-null   float64
18  Temperature (C)_2    31018 non-null   float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.5 MB

```

LOW CURRENT OCV AT 25 degree celcius

```
In [18]: low_curr_ocv_25=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\
```

```
In [19]: low_curr_ocv_25.head(10)
```

```
Out[19]:
```

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
0	1	3.003881	2012-09-05 09:50:31	3.003881	1	1	0.
1	2	8.004174	2012-09-05 09:50:36	5.000008	2	1	1.
2	3	13.004248	2012-09-05 09:50:41	10.000082	2	1	1.
3	4	18.004260	2012-09-05 09:50:46	15.000093	2	1	1.
4	5	23.005579	2012-09-05 09:50:51	20.001413	2	1	1.
5	6	28.005657	2012-09-05 09:50:56	25.001492	2	1	1.
6	7	33.011929	2012-09-05 09:51:01	30.007763	2	1	1.

2012-09-05 09:51:06 35.007785 2 1 1.

2012-09-05 09:51:11 40.007831 2 1 1.

2012-09-05 09:51:16 45.007870 2 1 1.

In [20]: `low_curr_ocv_25.describe()`

Out[20]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	32307.00000	32307.000000	32307	32307.000000	32307.000000
mean	16154.00000	85506.863991	2012-09-06 09:35:35.959884800	36302.546619	5.837713
min	1.00000	3.003881	2012-09-05 09:50:31	3.003881	1.000000
25%	8077.50000	43696.841249	2012-09-05 21:58:45.500000	15874.938811	5.000000
50%	16154.00000	84091.821468	2012-09-06 09:12:01	36076.586865	5.000000
75%	24230.50000	127800.461601	2012-09-06 21:20:30.500000	56272.784453	7.000000
max	32307.00000	171495.535618	2012-09-07 09:28:46	76592.209924	8.000000
std	9326.37191	48357.812231	NaN	22968.249246	1.217815

In [21]: `low_curr_ocv_25.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32307 entries, 0 to 32306
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            32307 non-null  int64
1   Test_Time(s)                          32307 non-null  float64
2   Date_Time                             32307 non-null  datetime64[ns]
3   Step_Time(s)                          32307 non-null  float64
4   Step_Index                            32307 non-null  int64
5   Cycle_Index                           32307 non-null  int64
6   Current(A)                            32307 non-null  float64
7   Voltage(V)                            32307 non-null  float64
8   Charge_Capacity(Ah)                   32307 non-null  float64
9   Discharge_Capacity(Ah)                 32307 non-null  float64
10  Charge_Energy(Wh)                      32307 non-null  float64
11  Discharge_Energy(Wh)                   32307 non-null  float64
```

```
12 dV/dt(V/s) 32307 non-null float64
13 Internal_Resistance(Ohm) 32307 non-null int64
14 Is_FC_Data 32307 non-null int64
15 AC_Impedance(Ohm) 32307 non-null int64
16 ACI_Phase_Angle(Deg) 32307 non-null int64
17 Temperature (C)_1 32307 non-null float64
dtypes: datetime64[ns](1), float64(10), int64(7)
memory usage: 4.4 MB
```

LOW CURRENT OCV AT 30 degree celcius

```
In [22]: low_curr_ocv_30=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\
```

```
In [23]: low_curr_ocv_30.head(10)
```

```
Out[23]:
```

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
0	1	3.010808	2012-06-25 10:58:13	3.010808	1	1	0.
1	2	3.041952	2012-06-25 10:58:13	0.030896	2	1	1.
2	3	8.057809	2012-06-25 10:58:18	5.015575	3	1	0.
3	4	13.065362	2012-06-25 10:58:23	10.023128	3	1	0.
4	5	18.072954	2012-06-25 10:58:28	15.030720	3	1	0.
5	6	23.080509	2012-06-25 10:58:33	20.038275	3	1	0.
6	7	28.088092	2012-06-25 10:58:38	25.045859	3	1	0.
7	8	33.095648	2012-06-25 10:58:43	30.053414	3	1	0.
8	9	38.103228	2012-06-25 10:58:48	35.060994	3	1	0.
9	10	43.110795	2012-06-25 10:58:53	40.068561	3	1	0.

In [24]: `low_curr_ocv_30.describe()`

Out[24]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	31150.000000	31150.000000	31150	31150.000000	31150.000000
mean	15575.500000	82879.119805	2012-06-26 09:59:30.199871488	37868.880320	5.967159
min	1.000000	3.010808	2012-06-25 10:58:13	0.030896	1.000000
25%	7788.250000	42289.364036	2012-06-25 22:43:00.24999872	18343.988166	5.000000
50%	15575.500000	81284.579150	2012-06-26 09:32:55.500000	37842.221660	5.000000
75%	23362.750000	123576.509714	2012-06-26 21:17:47.750000128	57340.455169	7.000000
max	31150.000000	165868.338984	2012-06-27 09:02:40	77269.004464	8.000000
std	8992.374779	46527.087207	NaN	22474.828502	1.040072

In [25]: `low_curr_ocv_30.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31150 entries, 0 to 31149
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            31150 non-null  int64
1   Test_Time(s)                          31150 non-null  float64
2   Date_Time                             31150 non-null  datetime64[ns]
3   Step_Time(s)                          31150 non-null  float64
4   Step_Index                            31150 non-null  int64
5   Cycle_Index                           31150 non-null  int64
6   Current(A)                            31150 non-null  float64
7   Voltage(V)                            31150 non-null  float64
8   Charge_Capacity(Ah)                   31150 non-null  float64
9   Discharge_Capacity(Ah)                31150 non-null  float64
10  Charge_Energy(Wh)                     31150 non-null  float64
11  Discharge_Energy(Wh)                  31150 non-null  float64
12  dV/dt(V/s)                            31150 non-null  float64
13  Internal_Resistance(Ohm)               31150 non-null  int64
14  Is_FC_Data                            31150 non-null  int64
15  AC_Impedance(Ohm)                     31150 non-null  int64
16  ACI_Phase_Angle(Deg)                  31150 non-null  int64
17  Temperature (C)_1                      31150 non-null  float64
18  Temperature (C)_2                      31150 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.5 MB
```

LOW CURRENT OCV AT 40 degree

ceicius

```
In [26]: low_curr_ocv_40=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\
```

```
low_curr_ocv_40.head(10)
```

```
Out[27]:
```

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
0	1	3.010283	2012-06-27 11:43:25	3.010283	1	1	0.
1	2	3.026131	2012-06-27 11:43:25	0.015609	2	1	1.
2	3	8.033910	2012-06-27 11:43:30	5.007549	3	1	0.
3	4	13.041500	2012-06-27 11:43:35	10.015140	3	1	0.
4	5	18.049071	2012-06-27 11:43:40	15.022711	3	1	0.
5	6	23.056643	2012-06-27 11:43:45	20.030282	3	1	0.
6	7	28.064215	2012-06-27 11:43:50	25.037854	3	1	0.
7	8	33.071785	2012-06-27 11:43:55	30.045425	3	1	0.
8	9	38.079286	2012-06-27 11:44:00	35.052926	3	1	0.
9	10	43.086932	2012-06-27 11:44:05	40.060571	3	1	0.

```
In [28]: low_curr_ocv_40.describe()
```

```
Out[28]:
```

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	31258.000000	31258.000000	31258	31258.000000	31258.000000
mean	15629.500000	83182.146533	2012-06-28 10:49:45.445390080	38403.674272	5.985124

min	1.000000	3.010283	2012-06-27 11:43:25	0.015609	1.000000
25%	7815.250000	42427.009169	2012-06-27 23:30:30.249999872	18823.463111	5.000000
50%	15629.500000	81557.428742	2012-06-28 10:22:40.500000	38388.046949	5.000000
75%	23443.750000	123985.717499	2012-06-28 22:09:49.750000128	57952.630847	7.000000
max	31258.000000	166414.916638	2012-06-29 09:56:59	77785.938692	8.000000
std	9023.551694	46653.945691	NaN	22568.608281	1.019973

In [29]:

low_curr_ocv_40.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31258 entries, 0 to 31257
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            31258 non-null  int64
1   Test_Time(s)                          31258 non-null  float64
2   Date_Time                             31258 non-null  datetime64[ns]
3   Step_Time(s)                          31258 non-null  float64
4   Step_Index                            31258 non-null  int64
5   Cycle_Index                           31258 non-null  int64
6   Current(A)                            31258 non-null  float64
7   Voltage(V)                            31258 non-null  float64
8   Charge_Capacity(Ah)                   31258 non-null  float64
9   Discharge_Capacity(Ah)                31258 non-null  float64
10  Charge_Energy(Wh)                      31258 non-null  float64
11  Discharge_Energy(Wh)                   31258 non-null  float64
12  dV/dt(V/s)                            31258 non-null  float64
13  Internal_Resistance(Ohm)               31258 non-null  int64
14  Is_FC_Data                             31258 non-null  int64
15  AC_Impedance(Ohm)                     31258 non-null  int64
16  ACI_Phase_Angle(Deg)                  31258 non-null  int64
17  Temperature (C)_1                      31258 non-null  float64
18  Temperature (C)_2                      31258 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.5 MB
```

LOW CURRENT OCV AT 50 degree celcius

In [30]:

low_curr_ocv_50=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\

In [31]:

low_curr_ocv_50.head(10)

Out[31]:

Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index	Cycle_Index	Cur
------------	--------------	-----------	--------------	------------	-------------	-----

0	1	3.008714	2012-07-02 10:52:00	3.008714	1	1	0.
1	2	3.020297	2012-07-02 10:52:00	0.000002	2	1	1.
2	3	8.037763	2012-07-02 10:52:05	5.007566	3	1	0.
3	4	13.045325	2012-07-02 10:52:10	10.015128	3	1	0.
4	5	18.052910	2012-07-02 10:52:15	15.022712	3	1	0.
5	6	23.060484	2012-07-02 10:52:20	20.030287	3	1	0.
6	7	28.068046	2012-07-02 10:52:25	25.037848	3	1	0.
7	8	33.075622	2012-07-02 10:52:30	30.045425	3	1	0.
8	9	38.083297	2012-07-02 10:52:35	35.053100	3	1	0.
9	10	43.090758	2012-07-02 10:52:40	40.060561	3	1	0.



In [32]: `low_curr_ocv_50.describe()`

Out[32]:

	Data_Point	Test_Time(s)	Date_Time	Step_Time(s)	Step_Index
count	31475.000000	31475.000000	31475	31475.000000	31475.000000
mean	15738.000000	83704.689997	2012-07-03 10:07:02.332041216	38368.988081	5.975473
min	1.000000	3.008714	2012-07-02 10:52:00	0.000002	1.000000
25%	7869.500000	42695.210585	2012-07-02 22:43:32.500000	18645.694362	5.000000
50%	15738.000000	82097.290906	2012-07-03 09:40:15	38347.986437	5.000000
75%	23606.500000	124794.372180	2012-07-03 21:31:52.500000	58047.774699	7.000000

max 31475.000000 167494.127364 2012-07-04 09:23:33 77843.159249 8.000000

std 9086.194198 46988.606063 NaN 22713.080086 1.035112

In [33]:

```
low_curr_ocv_50.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31475 entries, 0 to 31474
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Data_Point                            31475 non-null  int64
1   Test_Time(s)                         31475 non-null  float64
2   Date_Time                            31475 non-null  datetime64[ns]
3   Step_Time(s)                         31475 non-null  float64
4   Step_Index                           31475 non-null  int64
5   Cycle_Index                           31475 non-null  int64
6   Current(A)                           31475 non-null  float64
7   Voltage(V)                           31475 non-null  float64
8   Charge_Capacity(Ah)                  31475 non-null  float64
9   Discharge_Capacity(Ah)                31475 non-null  float64
10  Charge_Energy(Wh)                     31475 non-null  float64
11  Discharge_Energy(Wh)                   31475 non-null  float64
12  dV/dt(V/s)                           31475 non-null  float64
13  Internal_Resistance(Ohm)               31475 non-null  int64
14  Is_FC_Data                            31475 non-null  int64
15  AC_Impedance(Ohm)                     31475 non-null  int64
16  ACI_Phase_Angle(Deg)                   31475 non-null  int64
17  Temperature (C)_1                      31475 non-null  float64
18  Temperature (C)_2                      31475 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(7)
memory usage: 4.6 MB
```

DYNAMIC TEST PROFILES

Dynamic test profile at -10 degree celcius

In [34]:

```
dst_minus_10=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A12
```

In [35]:

```
dst_minus_10.head(10)
```

Out[35]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
0	3.003816	3.003816	1	0.000000	2.803019	-8.300812
1	8.004204	5.000077	2	-1.099373	2.109292	-8.145514
2	12.006524	9.002396	2	-1.099373	1.999416	-8.145514
3	17.013409	5.003358	3	0.000000	2.671906	-8.145514
4	22.015560	10.005508	3	0.000000	2.689757	-8.223158

5	27.017961	15.007909	3	0.000000	2.697759	-8.191391
6	32.018054	20.008003	3	0.000000	2.702376	-8.223158
7	37.026317	25.016265	3	0.000000	2.706993	-8.113752
8	42.030830	30.020778	3	0.000000	2.709763	-8.067879
9	47.039970	35.029918	3	0.000000	2.713148	-8.223158

In [36]: `dst_minus_10.describe()`

Out[36]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	19784.000000	1.978400e+04	19784.000000	19784.000000	19784.000000	1978
mean	20198.292967	8.537425e+02	15.393095	-0.322788	3.165822	-
std	9740.553386	1.314745e+03	6.824850	0.887062	0.304046	-
min	3.003816	3.849596e-07	1.000000	-3.849595	1.999416	-
25%	11169.170425	1.615422e+02	8.000000	-0.758130	2.985145	-
50%	20727.640514	3.380252e+02	16.000000	-0.008637	3.164809	-
75%	30207.416255	8.841617e+02	24.000000	0.134114	3.347782	-
max	35460.091077	7.315288e+03	27.000000	2.061261	3.814677	-

In [37]: `dst_minus_10.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19784 entries, 0 to 19783
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Test_Time(s)           19784 non-null  float64
1   Step_Time(s)           19784 non-null  float64
2   Step_Index             19784 non-null  int64
3   Current(A)             19784 non-null  float64
4   Voltage(V)             19784 non-null  float64
5   Temperature (C)_1      19784 non-null  float64
dtypes: float64(5), int64(1)
memory usage: 927.5 KB
```

Dynamic test profile at 0 degree celcius

In [38]: `dst_0=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A123 batte`

In [39]: `dst_0.head(10)`

Out[39]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
--	--------------	--------------	------------	------------	------------	-------------------

0	3.057882	3.005927	1	0.000000	2.846107	1.256174
1	8.065274	5.007092	2	-1.099185	2.468466	1.256174
2	13.072544	10.014362	2	-1.099373	2.403833	1.332009
3	18.077826	15.019644	2	-1.099373	2.333968	1.256174
4	23.078087	20.019905	2	-1.099373	2.250561	1.301033
5	28.082401	25.024220	2	-1.099560	2.134222	1.452703
6	31.764519	28.706338	2	-1.099373	1.999416	1.345893
7	36.774210	5.008228	3	0.000000	2.548488	1.332009
8	41.783489	10.017507	3	0.000000	2.566955	1.376867
9	46.792691	15.026708	3	0.000000	2.577727	1.332009

In [40]:

dst_0.describe()

Out[40]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	23248.000000	2.324800e+04	23248.000000	23248.000000	23248.000000	2324
mean	20466.229194	5.328935e+02	15.566027	-0.348093	3.165784	
std	10301.715583	6.231313e+02	6.763405	0.918652	0.259291	
min	3.057882	7.699193e-07	1.000000	-3.849219	1.961867	-
25%	10658.417985	1.537679e+02	8.000000	-0.829459	3.026156	
50%	20792.540701	3.166302e+02	16.000000	-0.191961	3.172349	
75%	30805.391681	6.670666e+02	24.000000	0.080958	3.293613	
max	36926.114253	4.328226e+03	27.000000	2.060698	3.728192	

In [41]:

dst_0.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23248 entries, 0 to 23247
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Test_Time(s)          23248 non-null  float64
1   Step_Time(s)          23248 non-null  float64
2   Step_Index            23248 non-null  int64
3   Current(A)            23248 non-null  float64
4   Voltage(V)            23248 non-null  float64
5   Temperature (C)_1     23248 non-null  float64
dtypes: float64(5), int64(1)
memory usage: 1.1 MB
```

Dynamic test profile at 10 degree celcius

In [42]:

dst_10=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A123 batt

In [43]:

dst_10.head(10)

Out[43]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
0	3.008291	3.008291	1	0.000000	2.968602	9.646121
1	8.015758	5.007161	2	-1.099185	2.729152	9.690903
2	13.018937	10.010341	2	-1.099373	2.703299	9.598107
3	18.027239	15.018642	2	-1.099373	2.678369	9.797547
4	23.031320	20.022724	2	-1.099373	2.654363	9.766615
5	28.035736	25.027140	2	-1.099373	2.629125	9.721836
6	33.039112	30.030516	2	-1.099373	2.603272	9.797547
7	38.047386	35.038790	2	-1.099373	2.576496	9.690903
8	43.053577	40.044981	2	-1.099373	2.547873	9.918034
9	48.062643	45.054046	2	-1.099373	2.515864	9.873258

In [44]:

dst_10.describe()

Out[44]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	23881.000000	2.388100e+04	23881.000000	23881.000000	23881.000000	2388
mean	19919.260228	4.755559e+02	15.518613	-0.356659	3.190354	1
std	10203.319502	4.779053e+02	6.705619	0.932044	0.224014	
min	3.008291	7.699193e-07	1.000000	-3.849407	1.987413	
25%	10199.540085	1.497195e+02	8.000000	-0.848430	3.077862	1
50%	20191.833861	3.095074e+02	16.000000	-0.243238	3.196048	1
75%	30156.128886	5.997450e+02	24.000000	0.072130	3.302231	1
max	36435.385542	2.681824e+03	27.000000	2.061261	3.699877	1

In [45]:

dst_10.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23881 entries, 0 to 23880
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Test_Time(s)          23881 non-null float64
1   Step_Time(s)          23881 non-null float64
2   Step_Index            23881 non-null int64
3   Current(A)            23881 non-null float64
4   Voltage(V)            23881 non-null float64
```

```
5 Temperature (C)_1 23881 non-null float64
dtypes: float64(5), int64(1)
memory usage: 1.1 MB
```

Dynamic test profile at 20 degree celcius

```
In [46]: dst_20=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A123 batt
```

```
In [47]: dst_20.head(10)
```

```
Out[47]:
```

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
0	3.008332	3.008332	1	0.000000	3.108637	18.427523
1	8.014952	5.006227	2	-1.099776	2.891659	18.444551
2	13.018221	10.009497	2	-1.099776	2.877501	18.427523
3	18.022437	15.013712	2	-1.099957	2.864575	18.489090
4	23.026696	20.017971	2	-1.099776	2.851957	18.382978
5	28.035930	25.027205	2	-1.099957	2.839954	18.458305
6	33.045322	30.036597	2	-1.099776	2.827950	18.365955
7	38.051520	35.042795	2	-1.099957	2.815948	18.427523
8	43.066605	40.057881	2	-1.099776	2.803944	18.352194
9	48.077106	45.068381	2	-1.099776	2.791941	18.321405

```
In [48]: dst_20.describe()
```

```
Out[48]:
```

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	24992.000000	2.499200e+04	24992.000000	24992.000000	24992.000000	2499
mean	20478.506604	4.711195e+02	15.573944	-0.362114	3.195718	2
std	10524.199546	4.800529e+02	6.722357	0.935984	0.229664	
min	3.008332	7.699193e-07	1.000000	-3.849713	1.898483	1
25%	10446.853748	1.524163e+02	8.000000	-0.862705	3.091094	1
50%	20715.938621	3.099327e+02	16.000000	-0.279794	3.208663	1
75%	30962.624857	5.888735e+02	24.000000	0.063971	3.315767	2
max	37521.292897	2.867548e+03	27.000000	2.061054	3.732181	2

```
In [49]: dst_20.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 24992 entries, 0 to 24991

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	Test_Time(s)	24992 non-null	float64
1	Step_Time(s)	24992 non-null	float64
2	Step_Index	24992 non-null	int64
3	Current(A)	24992 non-null	float64
4	Voltage(V)	24992 non-null	float64
5	Temperature (C)_1	24992 non-null	float64

dtypes: float64(5), int64(1)

memory usage: 1.1 MB

Dynamic test profile at 25 degree celcius

```
In [50]: dst_25=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A123 batt
```

```
In [51]: dst_25.head(10)
```

```
Out[51]:
```

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
0	3.053504	3.000204	1	0.000000	2.766393	26.788187
1	8.061154	5.007273	2	-1.099373	2.501398	26.818743
2	13.062293	10.008411	2	-1.099185	2.435226	26.818743
3	18.071620	15.017739	2	-1.099373	2.340432	26.713451
4	23.073773	20.019892	2	-1.099185	2.115448	26.788187
5	24.301853	21.247971	2	-1.099373	1.998801	26.893473
6	29.307341	5.004141	3	0.000000	2.460464	26.893473
7	34.311587	10.008388	3	0.000000	2.474314	26.713451
8	39.319818	15.016618	3	0.000000	2.482316	26.818743
9	44.325095	20.021896	3	0.000000	2.486933	26.937641

```
In [52]: dst_25.describe()
```

```
Out[52]:
```

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	24469.000000	2.446900e+04	24469.000000	24469.000000	24469.000000	2446
mean	19726.353242	4.718362e+02	15.557767	-0.363447	3.204312	2
std	10192.971787	5.018513e+02	6.710170	0.942908	0.211759	
min	3.053504	3.849596e-07	1.000000	-3.849407	1.937861	2
25%	10017.592323	1.497245e+02	8.000000	-0.867589	3.110487	2
50%	19958.708627	3.061075e+02	16.000000	-0.286440	3.214515	2
75%	29860.944817	5.805784e+02	24.000000	0.060672	3.311772	2

max 36294.795004 3.070756e+03 27.000000 2.061261 3.699877 2

In [53]:

```
dst_25.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24469 entries, 0 to 24468
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Test_Time(s)          24469 non-null  float64
1   Step_Time(s)          24469 non-null  float64
2   Step_Index            24469 non-null  int64
3   Current(A)            24469 non-null  float64
4   Voltage(V)            24469 non-null  float64
5   Temperature (C)_1     24469 non-null  float64
dtypes: float64(5), int64(1)
memory usage: 1.1 MB
```

Dynamic test profile at 30 degree celcius

In [54]:

```
dst_30=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A123 batt
```

In [55]:

```
dst_30.head(10)
```

Out[55]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
0	3.004871	3.004871	1	0.000000	3.075400	31.391201
1	8.011347	5.006164	2	-1.099373	2.888580	31.242373
2	13.017574	10.012391	2	-1.099373	2.876269	31.391201
3	18.023831	15.018647	2	-1.099373	2.864882	31.421654
4	23.027999	20.022816	2	-1.099373	2.853494	31.496058
5	28.032410	25.027227	2	-1.099373	2.842414	31.421654
6	33.039675	30.034492	2	-1.099373	2.831334	31.421654
7	38.041781	35.036597	2	-1.099373	2.819946	31.465605
8	43.043244	40.038061	2	-1.099373	2.808558	31.465605
9	48.043298	45.038114	2	-1.099560	2.796555	31.391201

In [56]:

```
dst_30.describe()
```

Out[56]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	24598.000000	2.459800e+04	24598.000000	24598.000000	24598.000000	2459
mean	19621.906470	4.721762e+02	15.567973	-0.364780	3.206956	3

std	10170.231343	5.126144e+02	6.716462	0.942931	0.203930	
min	3.004871	3.849596e-07	1.000000	-3.849407	1.949556	3
25%	9938.020160	1.497872e+02	8.000000	-0.871909	3.116026	3
50%	19831.893588	3.053628e+02	16.000000	-0.297146	3.217592	3
75%	29711.598095	5.757773e+02	24.000000	0.049966	3.312695	3
max	36178.975170	3.143676e+03	27.000000	2.061261	3.696491	3

In [57]:

dst_30.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24598 entries, 0 to 24597
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Test_Time(s)          24598 non-null  float64
1   Step_Time(s)          24598 non-null  float64
2   Step_Index            24598 non-null  int64
3   Current(A)            24598 non-null  float64
4   Voltage(V)            24598 non-null  float64
5   Temperature (C)_1     24598 non-null  float64
dtypes: float64(5), int64(1)
memory usage: 1.1 MB
```

Dynamic test profile at 50 degree celcius

In [58]:

dst_50=pd.read_excel(r"C:\Users\jaiku\PycharmProjects\Assignments\A123 batt

In [59]:

dst_50.head(10)

Out[59]:

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Temperature (C)_1
0	3.007451	3.007451	1	0.000000	2.775626	50.789497
1	8.012196	5.004451	2	-1.099373	2.564185	50.789497
2	13.012202	10.004457	2	-1.099373	2.523558	50.789497
3	18.014646	15.006900	2	-1.099373	2.473083	50.746269
4	23.021725	20.013979	2	-1.099373	2.404141	50.879433
5	28.021949	25.014203	2	-1.099185	2.257025	50.952637
6	30.994169	27.986424	2	-1.099185	1.999724	50.746269
7	35.996811	5.001098	3	0.000000	2.367208	50.836208
8	41.000069	10.004355	3	0.000000	2.385059	50.866188
9	46.000322	15.004608	3	0.000000	2.393369	50.819477

```
In [60]: dst_50.describe()
```

```
Out[60]:
```

	Test_Time(s)	Step_Time(s)	Step_Index	Current(A)	Voltage(V)	Tem
count	24763.000000	2.476300e+04	24763.000000	24763.000000	24763.000000	2476
mean	19197.121372	4.804346e+02	15.652263	-0.367141	3.214068	5
std	10034.610605	5.380568e+02	6.695618	0.947778	0.195042	
min	3.007451	3.849596e-07	1.000000	-3.849407	1.926166	5
25%	9617.456118	1.515973e+02	8.000000	-0.880455	3.126183	5
50%	19349.539252	3.072322e+02	16.000000	-0.311609	3.226826	5
75%	29076.097827	5.734703e+02	24.000000	0.034939	3.320697	5
max	35582.345606	3.301138e+03	27.000000	2.061261	3.709726	5

```
In [61]: dst_50.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 24763 entries, 0 to 24762
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Test_Time(s)          24763 non-null  float64
1   Step_Time(s)          24763 non-null  float64
2   Step_Index            24763 non-null  int64
3   Current(A)            24763 non-null  float64
4   Voltage(V)            24763 non-null  float64
```